Introduction to REST API

Overview



What is REST API?

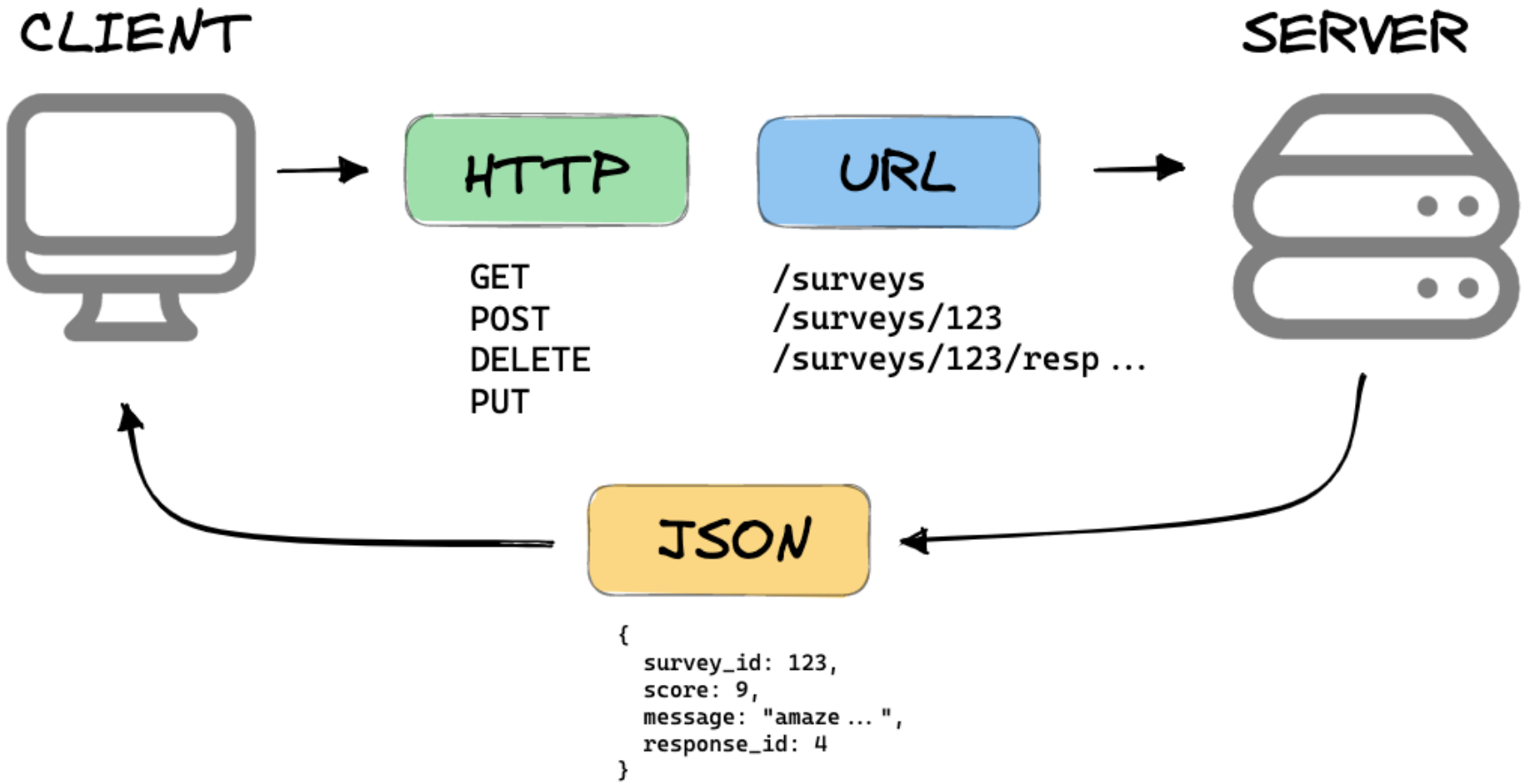
History of REST API

Characteristics of REST API

What is REST API?

- **REST** stands for **Representational State Transfer**.
- It is an architectural style for designing networked applications, and RESTful APIs (Application Programming Interfaces) are a way to implement that style.
- It provides a standardized way for clients (such as web browsers, mobile apps, or other servers) to interact with server-side resources.
- They leveraged existing HTTP protocols and methods like **GET**, **POST**, **PUT**, and **DELETE** to interact with resources.
- They provide a simple and lightweight way to create and consume web services that can be easily understood and integrated by various clients and platforms.

WHAT IS A REST API?



History of REST API

- It was introduced by **Roy Fielding** in his 2000 doctoral dissertation titled "Architectural Styles and the Design of Network-based Software Architectures."
- Which outlined principles for building scalable and stateless web applications.
- RESTful APIs gained popularity for their simplicity, scalability, and ease of use.
- Major companies and platforms started adopting RESTful APIs as a way to provide programmatic access to their services.
- To ensure consistency and ease of use, standards for API design and documentation emerged. Tools like Swagger (now known as OpenAPI) and RAML helped developers define and document REST APIs more comprehensively, making it easier for others to understand and use the APIs.
- While RESTful APIs continued to be widely used, GraphQL, introduced by Facebook, gained popularity as an alternative API approach.

Characteristics of REST API

- **Stateless**: The server does not store any client state between requests. Each request from the client must contain all the necessary information to understand and process it.
- **Client-Server Architecture:** The client and server are separate entities that communicate over the network. The client is responsible for the user interface, while the server handles the business logic and data storage.
- **Uniform Interface:** REST APIs have a consistent and uniform interface, which typically includes the use of HTTP methods (such as GET, POST, PUT, DELETE) for different operations on resources.

- **Resource-Oriented:** REST APIs are based on the concept of resources, which can be any entity or object that needs to be exposed and manipulated. Resources are identified by unique URIs (Uniform Resource Identifiers).
- **Support for Multiple Formats:** REST APIs can support different data formats such as **JSON (JavaScript Object Notation)** and XML (eXtensible Markup Language) for representing and transferring data.

References

https://aws.amazon.com/what-is/restful-api/

https://www.ibm.com/topics/rest-apis

https://restfulapi.net/

https://www.techtarget.com/searchapparchitecture/definition/RESTful-API

https://blog.hubspot.com/website/what-is-rest-api

https://www.javatpoint.com/what-is-rest-api

https://www.smashingmagazine.com/2018/01/understanding-using-rest-api/